git hub steps
1. Open the repo for which you have the access
2. To clone/copy the repo in to your local machines (only once)
git clone https://github.com/sunbeam-modular/CJ-O-17.git
3. To get the new changes added in the repo
git pull

OOP
Language Fundamentals
Widening/narrowing

functions() -> methods()
syntax
Rules

1. JDK -> java Development Kit
2. IDE -> STS / Intellij

javac Program.java
java Program

When .java file is compiled it will create a .class file
As many no classes exists in that .java file those many .class files will be created

JVM -> Java Virtual Machine

Naming Convention
1. Pascal Case
    - the first letter of ever word should be capital
    - Employee,Student, PrintStream,NullPointerException
    - class,interface,enum
2. Camel Case
    - the first letter of every word should be capital except the first word
    - salary,totalSalary,calculateTotalSalary
    - local variable, fields,methods,paramaters
3. Packages
    - every word needs to be in small case
4. Final/constat variables
    - Every word needs to be in uppercase
    - PI

java 8 documentation link
https://docs.oracle.com/javase/8/docs/api/

- name of public class and the name of file in which it is declared must be same
- in a single .java file their can be only 1 public class
- in a single .java file their can be multiple non public classes

Datatypes

- datatypes defines 3 things
1. nature -> type of data that can be stored (Whole/Fractional numbers, Letter, Word)
2. memory -> The space required to store that data
3. operations -> The difeerent operations that can be carried out on the given data

types of Datatypes in java

1. Primitive (Value Type)
    a. Boolean
        - boolean (true or false) - 1 bit
    b. Character
        - char (unicode) - 2 bytes
    c. Integral
        - byte (numbers) - 1 byte
        - short - 2 bytes
        - int - 4 bytes
        - long - 8 bytes
    4. Floating-point
        - float - 4 bytes
        - double - 8 bytes
2. Non Primitive (Reference Type)
    1. Class
    2. Array
    3. Interface
    4. Enum

in_punch -> in_time(hr ,min)
out_punch -> out_time (hr,min)

i_hr = 10 i_min = 30
o_hr = 17 o_min = 45
1 employee

e2i_hr = 10 e2i_min = 30
e2o_hr = 17 e2o_min = 45
1 employee

Why we need the user defined datatype?

Anil	Mukesh	Ramesh
int day;	int day;	int day;
int month;	int month;	int month;
int year;	int year;	int year;

```
class Time{
    int hr;
    int min;
}

Time e1_in,e1_out;
Time e2_in,e2_out;
```

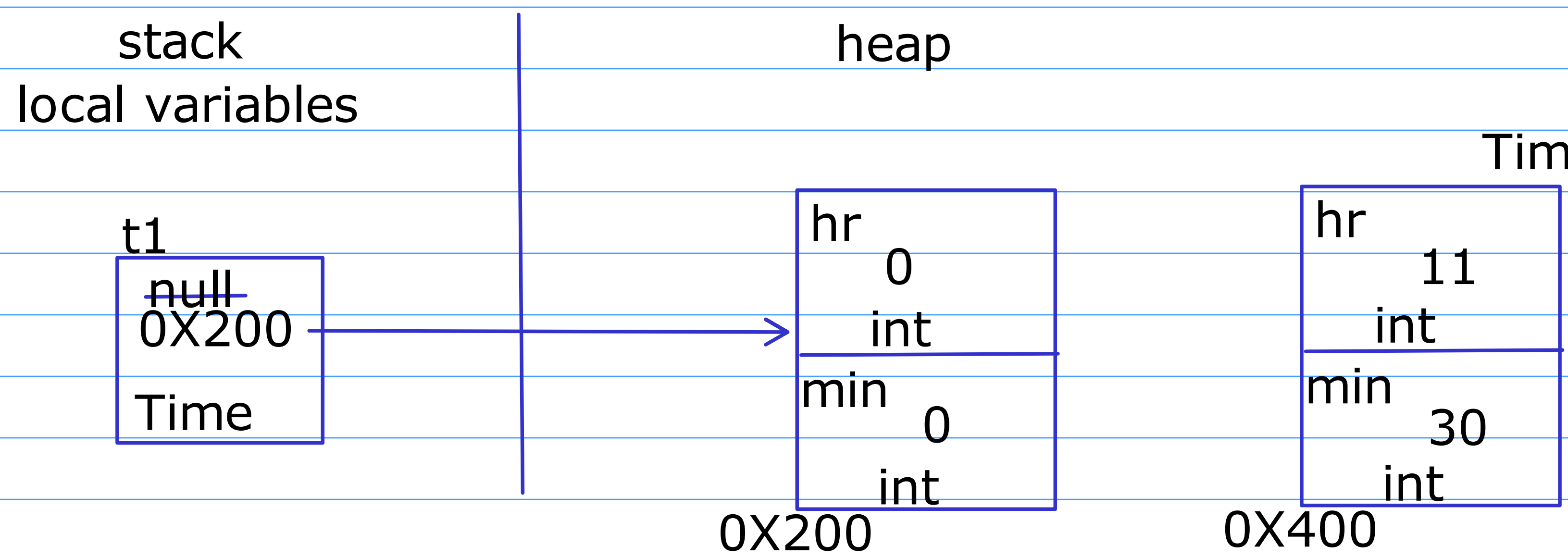
variable

datatype identifier
int n1;

n1
10
int
4 bytes
0X200

class
- It is a logical entity
- It binds the data and code together
- It is an example of encapsulation
- It is also called as blueprint of an object
- It consists of Fields and methods

variable
field
reference



Time t1 = new Time();

new Time();

- All objects in java are created dynamically.
- once their external references are deleted then those objects are also removed from the memory(heap) by the garbage collector

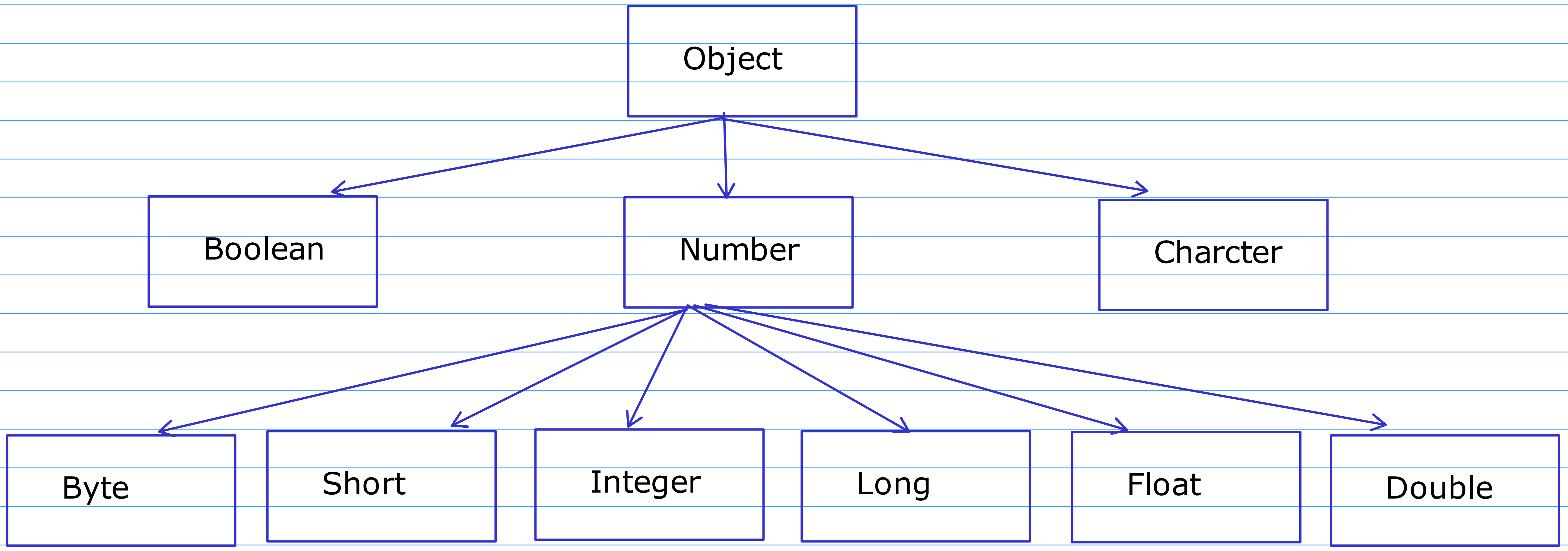
Object
- It is physical entity
- It is also called as instance of a class
- It defines 3 things
1. State
    - It is represented by the fields of the class
2. Behaviour
    - Methods of the class represents the behvaiour
3. Identity
    - Unique field can be used as an identity of an object
    - If no unique field exists then address can be used as an identity

```
class BankAccount{
int accno;
}

class Student{
int rollno;
}
```

Wrapper class

For every primitive type , java have defined a class for it called as a Wrapper class



1. To use the primitive types as reference types

    the prmitive types we cannot use it in collections, it has to be a reference type

2. These wrapper classes provides some helper methods